

Ad-hoc model licensing creates systemic legal risk

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Authors

Moming Duan*, Mingzhe Du*, Rui Zhao, Mengying Wang, Yinghui Wu, Nigel Shadbolt, Bingsheng He

Affiliations

NUS IDS; Oxford CS; CWRU CDS; NUS SoC

Contact

moming@nus.edu.sg • * Equal contribution

Project

ModelGo Licenses (MGLs): <https://www.modelgo.li/>

Widespread mismatch, proliferation, and conflicts persist

Implicit dependencies magnify compliance uncertainty

- Non-ML licenses misfit model realities.
- Derivatives republished under new terms.
- Incompatible clauses collide downstream.
- Implicit reuse spreads restrictions silently.

HF dependency graph

 Graph visualization

Finetune/adapter dominate; merges knit dense nets.

Mismatch exceeds 75% despite recent improvements

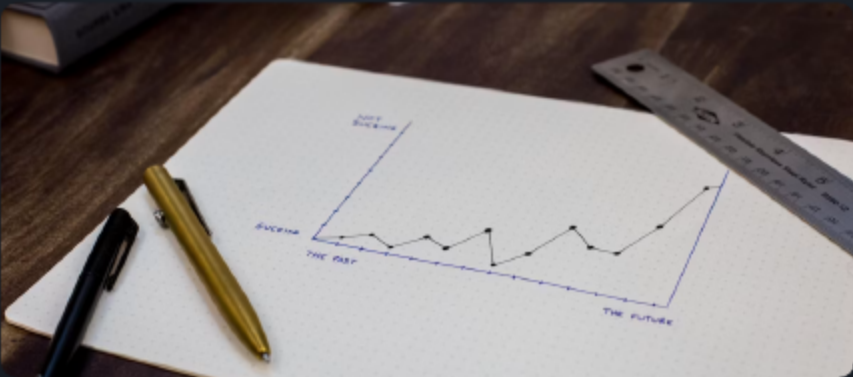
Null/custom terms rise; RAIL dips increase mismatch

License preferences



Apache/MIT lead; many null/custom licenses.

Adoption trends



Mismatch falls with RAILS; rises post-07/2024.

Behavioral terms drive most observed conflicts

2.2% conflicted models; Llama2/3 dominate

Derivative licensing pattern



53% retain; 28.7% none; 12.8% switch OSS.

Conflicts and spread



Exclusivity and ND fuel incompatibilities.

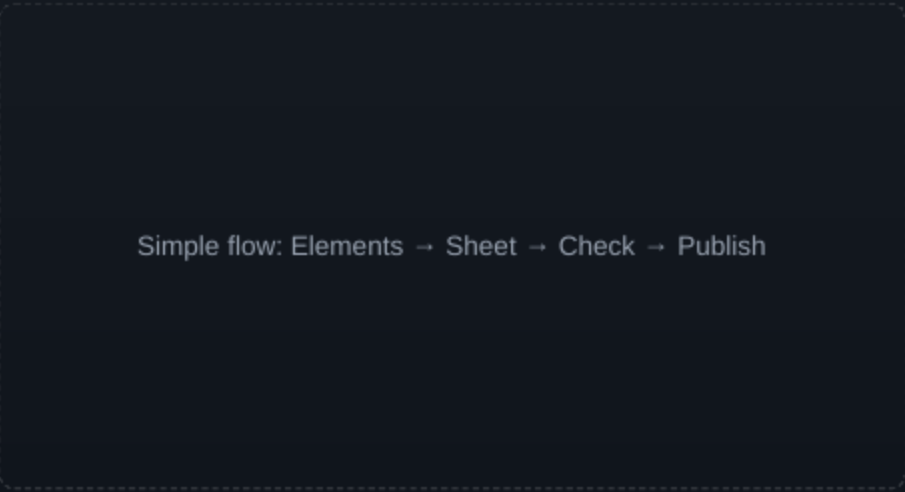
Our most surprising finding

Copyleft-style behavior terms outperform classic copyleft at causing conflicts.

- Why unexpected: OSS copyleft drives most issues.
- But model behavior clauses dominate conflicts.

We provide MGLs and a practical compliance workflow

- 1 Select MGL elements (BY/NC/ND/RAI/OS).
- 2 Fill model sheet fields clearly.
- 3 Map dependencies and licenses.
- 4 Run conflict checks automatically.
- 5 Resolve flags or adjust elements.
- 6 Publish sheet with the model.
- 7 Monitor derivatives downstream.
- 8 Iterate with ablations/limits.



MGLs clarify derivatives, outputs, and remote use

- License matrix: clarity dimensions covered.
- Clauses: ownership and patent grants.
- Model sheet: human-readable summary.
- Conflict-check: rules from license text.

Clarity matrix snapshot

Matrix grid

MGLs score high, reusable across projects.

Workflow, examples, and known limitations

- Workflows: finetune, adapter, quantize, merge.
- Examples: Llama3.2 derivative licensing.
- Limitations: self-reported dependencies.
- Ablations: qualified subsets sensitivity.

Process sketch



Path-specific checks per reuse method.

Worked example: MG-BY-OS-2.0 compliance decision

Input model metadata

- Base: Llama-3.2-3B-Instruct.
- Reuse: finetune, merge.
- Current: no license.
- Outputs: text generation.

Model sheet outcome

- Elements: BY, OS.
- Derivatives: allowed.
- Outputs: allowed.
- Decision: noncompliant until license added.

Takeaway: add MG-BY-OS-2.0; recheck merges for exclusivity.

MGLs enable clearer, safer reuse across the ML supply chain

Evidence: high mismatch, behavior-driven conflicts, workable checks

- Provide reusable, model-focused licenses.
- Support sheets and automated checks.
- Reduce conflicts and silent violations.
- Next: standards, metadata, and tooling.